

Ironwood, the core

how Zcash verifies a shielded transfer without seeing it

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What a validator checks

1. the coin **exists** — look it up
2. the spend is **authorised** — script and signature
3. it is not spent **twice** — delete it
4. value is **conserved** — $\sum \text{in} \geq \sum \text{out}$

Every check reads the transaction in the clear: which coin, whose key, how much.

Privacy, precisely

Hide the sender, the receiver and the amount — while a stranger still runs all four checks.

check	mechanism
exists	a commitment in a Merkle tree, membership proved in zero knowledge
not twice	a nullifier: a keyed serial number, revealed once
conserved	homomorphic value commitments and a binding signature
authorised	a proof of key knowledge and a re-randomised signature

One zero-knowledge statement per Action carries all four; one proof per bundle.

Anchors

A group and a hard problem

$$\mathcal{E}(\mathbb{F}_p) : y^2 = x^3 + 5 \quad \text{over } \mathbb{F}_p, \quad p \approx 2^{254}, \quad |\mathcal{E}| = q \text{ prime}$$

$$[a]G := \underbrace{G + G + \cdots + G}_{a \text{ times}} \quad \text{given } G \text{ and } [a]G, \text{ find } a : \text{ infeasible}$$

- Pallas: points over $\mathbb{F}_p$, scalars in $\mathbb{F}_q$.
- Vesta: the same curve with p and q swapped — a cycle.
- Everything below is built from $[a]G$ and the discrete log.

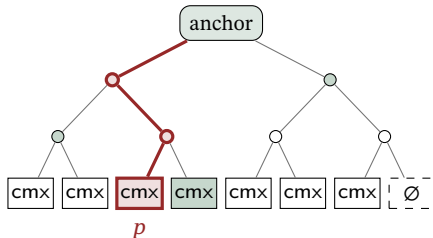
Commitments

$$C(v, r) := [v] V + [r] R \quad V, R \text{ independent points}$$

- **Hiding:** r uniform, so C is a uniform point; v is invisible.
- **Binding:** two openings $(v, r) \neq (v', r')$ of one C give $\log_R V = (r' - r)/(v - v')$.
- **Homomorphic:** $C(v_1, r_1) + C(v_2, r_2) = C(v_1 + v_2, r_1 + r_2)$.

A sealed envelope that can be added to other sealed envelopes.

Hashes and Merkle trees



- Hash H collision resistant: nobody can find two inputs with one output.
- The root commits to every leaf.
- Membership of leaf p : its 32 siblings, hashed upward, reach the root.
- Append only: old roots stay valid forever.

A sigma protocol: Schnorr identification

Prover knows x with $X = [x] G$.

Prover

Verifier

$R = [k] G, k \text{ random}$

$c \text{ random}$

$s = k + c x$

Verifier checks $[s] G \stackrel{?}{=} R + [c] X$

Why it proves, and why it hides

Special soundness. Two accepting transcripts with the same R :

$$(R, c, s), (R, c', s') \implies x = \frac{s - s'}{c - c'}$$

whoever answers two challenges knows x .

Zero knowledge. A simulator with no x picks c and s first:

$$R := [s] G - [c] X$$

the transcript has exactly the real distribution. The verifier learns nothing it could not have made itself.

Fiat-Shamir: the verifier becomes a hash

$$c := H(X, R, m)$$

- The prover cannot pick R after c : c depends on R .
- The pair (R, s) is now a signature on m — Schnorr.
- **sigma protocol + Fiat-Shamir = Schnorr signature.**
- Zcash's RedPallas is Schnorr on Pallas, with one addition: the key can be re-randomised (later).

Zero-knowledge proofs, in general

A relation $\mathcal{R}(x, w)$: public statement x , private witness w .

$$\text{Prove}(x, w) \rightarrow \pi \qquad \text{Verify}(x, \pi) \in \{0, 1\}$$

- **Complete:** an honest π verifies.
- **Knowledge sound:** from any convincing prover an extractor pulls a w with $\mathcal{R}(x, w)$.
- **Zero knowledge:** a simulator given only x reproduces π .
- **SNARK:** π short (logarithmic) and non-interactive (Fiat–Shamir).

Schnorr is the proof for $\mathcal{R} = \{(X, x) : X = [x] G\}$.

A polynomial cannot lie everywhere

$$D \neq 0, \deg D \leq d \implies \Pr_{z \leftarrow \mathbb{F}}[D(z) = 0] \leq \frac{d}{|\mathbb{F}|}$$

A nonzero polynomial has at most d roots. Fix D first, draw z second: one random probe catches it.

With $|\mathbb{F}| \approx 2^{254}$ and $d \approx 2^{14}$ the odds of escape are about 2^{-240} .

The core of Orchard

A note

$$n = (d, pk_d, v, \rho, \psi, rcm)$$

$$cm = \text{NoteCommit}_{rcm}(g_d, pk_d, v, \rho, \psi), \quad cmx = \text{Extract}(cm) \in \mathbb{F}_p$$

(d, pk_d) the recipient's diversified address

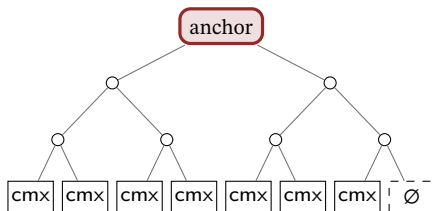
v the value, $0 \leq v < 2^{64}$

ρ, ψ uniqueness seed and nullifier salt

rcm the commitment trapdoor

On chain: cmx and a ciphertext. Nothing else about the note.

The chain's shielded state



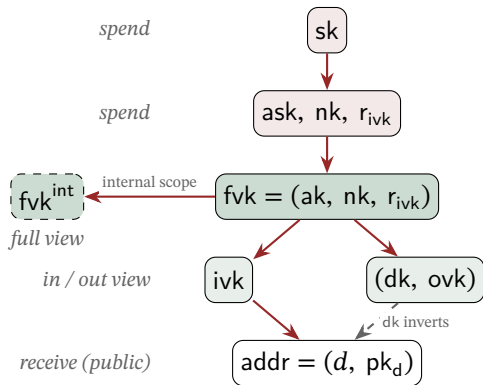
- **The commitment tree:** depth 32, append only; every block's final root is an *anchor*.
- **The nullifier set:** the spent half, growing; a nullifier names no leaf.
- A spend proves membership under some anchor and reveals a nullifier.

The nullifier

$$\text{nf} := \text{Extract}([F_{\text{nk}}(\rho) + \psi] G^{\text{nf}} + \text{cm})$$

- Keyed by nk , the owner's nullifier key; F_{nk} is Poseidon, a hash the circuit evaluates natively.
- Seeded by ρ , salted by ψ , bound to cm .
- Chaining $\rho_{\text{new}} := \text{nf}_{\text{old}}$: every note inherits the nullifier of the note spent beside it, so ρ is unique.
- Unique, and unlinkable to cm without nk .

Keys



$ask, nk, r_{ivk} := \text{PRFexpand}_{sk}(06), (07), (08)$

$ak := [ask] G^{\text{SpendAuth}}$

$ivk := \text{Commit}_{r_{ivk}}^{ivk}(\text{Extract}(ak), nk)$
 $(dk, ovk) := \text{PRFexpand}_{r_{ivk}}(82 \parallel \text{Extract}(ak) \parallel nk)$

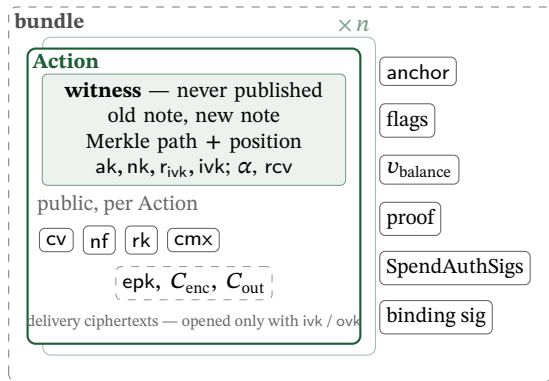
$d := \text{FF1}_{dk}(j), g_d := \text{GroupHash}(d),$
 $pk_d := [ivk] g_d$

Diversified addresses

$$d \in \{0, 1\}^{88}, \quad g_d := \text{GroupHash}(d), \quad pk_d := [ivk] g_d$$

- Some 2^{88} addresses per key, all paid into one wallet.
- Unlinkable to each other, even by the payer: two addresses share ivk only as a discrete log.
- One ivk detects every payment to every address.

The Action



- One Action spends one note and creates one note.
- A dummy note pads either side; every Action looks alike.
- One proof per bundle covers all its Actions.

The statement

check	clause	the proof establishes
exists	A3	a Merkle path from cm_{old} to the anchor
not twice	A5	the nullifier nf is this note's serial, keyed by nk
conserved	A4	the commitment cv hides $v_{old} - v_{new}$, both $< 2^{64}$
authorised	A6, A7	the key rk disguises ak ; ak, nk sit behind the address
well-formed	A1, A2	both commitments open to notes; $\rho_{new} = nf_{old}$
gated	A8–A10	spends, outputs, cross-address payments switched by flags

The witness: both notes, the path and position, the keys, α , rcv .

The one check the proof cannot make: has this nf been seen before.

Conservation without amounts

$$\begin{aligned} \text{cv} &= [v_{\text{old}} - v_{\text{new}}] V + [\text{rcv}] R && \text{per Action} \\ \text{bvk} &:= \sum_i \text{cv}_i - [v_{\text{balance}}] V = [\sum_i \text{rcv}_i] R && \text{iff the books balance} \end{aligned}$$

- The balance v_{balance} is public: the net flow out of the pool; for a fully shielded transfer, the fee.
- The binding signature is Schnorr under bvk on base R : knowing its secret key means knowing $\sum \text{rcv}$, which exists only when the amounts cancel.

Authorisation without a fixed key

$$rk := ak + [\alpha] G^{\text{SpendAuth}}, \quad rsk := ask + \alpha$$

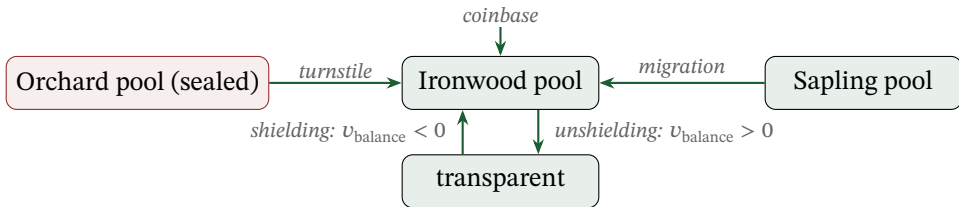
- One ak per wallet; publishing it would stamp every spend.
- Fresh α per spend: rk is a uniform point, unlinkable.
- The signature is Schnorr under rk ; the circuit proves rk was formed from the ak behind the spent note's address.

Delivery

$$\begin{aligned} \text{epk} &:= [\text{esk}] g_d, & s &:= [\text{esk}] \text{pk}_d = [\text{ivk}] \text{epk}, & K &:= \text{KDF}(s, \text{epk}) \\ C_{\text{enc}} &:= \text{AEAD}_K(0x03, d, v, \text{rseed}, \text{memo}) \end{aligned}$$

- One-shot Diffie–Hellman to the diversified base, then ChaCha20-Poly1305.
- No address on chain: the recipient trial-decrypts every output with ivk. A wrong key is rejected by the tag.
- The ciphertext C_{out} is the sender's own copy, under ovk.

The doors



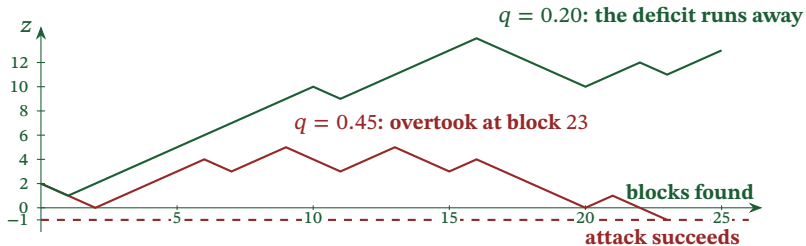
Every crossing is public in amount. Inside a pool, nothing is. Each pool's total is public and may never go negative.

What consensus adds

- A nullifier must not repeat within a transaction or across the valid chain.
- Transactions apply in block order: check $nf \notin \text{set}$, insert nf , append cmx .
- An anchor must be the final root of an earlier block — no spending an unmined note.
- A reorganisation rolls back tree, nullifier set and pool totals together.

The same shape as the UTXO set: an ordered, stateful check — over serial numbers that name no output.

Why one fork wins



Honest lead z ; each block moves it $+1$ with probability p , -1 with probability q .

$$a_z = \Pr[\text{lead ever reaches } -1] = (q/p)^{z+1} \quad (q < p)$$

Each block of deficit multiplies the chance by q/p : two forks cannot compete for long, and one quickly wins.

How Halo 2 proves

The claim becomes a table

“I know x with $x^3 + x + 5 = 35$ ” over $\mathbb{F}_{97}$. One operation per row, one gate equation on every row:

$$q_M a b + q_L a + q_R b + q_O c + q_C + \text{PI} = 0$$

row	advice			fixed					instance	
	a	b	c	q_M	q_L	q_R	q_O	q_C	PI	
0	3	3	9	1	0	0	-1	0	0	$x \cdot x = x^2$
1	9	3	27	1	0	0	-1	0	0	$x^2 \cdot x = x^3$
2	27	3	30	0	1	1	-1	0	0	$x^3 + x = s$
3	30	0	0	0	1	0	0	5	-35	$s + 5 = 35$

Advice: the prover's secret trace. Fixed: the circuit. Instance: the public 35. Copy constraints wire the rows; range checks are lookups into a table.

Rows become polynomials

Name the rows by roots of unity: $\omega = 22$, $H = \{1, 22, 96, 75\}$, $\omega^4 = 1$.

$$a(1) = 3, a(22) = 9, a(96) = 27, a(75) = 30 \implies a(X) = 90 + 61X + 22X^2 + 24X^3$$

$$G(X) := q_M ab + q_L a + q_R b + q_O c + q_C + \text{PI}$$

$$Z_H(X) := \prod_{h \in H} (X - h) = X^4 - 1$$

$$\text{every gate holds} \iff G \text{ vanishes on } H \iff Z_H \mid G \iff G = t \cdot Z_H$$

A correct computation is a polynomial identity. A wrong one cannot be: the lie $x^2 = 10$ in row 0 leaves a remainder $24(1 + X + X^2 + X^3) \neq 0$.

One random point

1. The prover *commits* to a, b, c and t .
2. The challenge $z := H(\text{transcript})$ — Fiat–Shamir.
3. The prover opens $a(z), b(z), c(z), t(z)$.
4. The verifier checks one equation:

$$G(z) \stackrel{?}{=} t(z) Z_H(z)$$

Honest, $z = 20$: $G(20) = 75 = 67 \cdot 46$. The row-0 lie: $G^*(20) = 20 \neq 64$.

A false identity is frozen before z is drawn; it survives only if z is one of its few roots.

Committing to a polynomial

$$P := \langle \vec{a}, \vec{G} \rangle + [r]H, \quad \vec{a} = (a_0, \dots, a_{d-1}), \quad \vec{G} \text{ independent points}$$
$$a(z) = \langle \vec{a}, \vec{b} \rangle, \quad \vec{b} := (1, z, z^2, \dots, z^{d-1})$$

- A Pedersen commitment to the coefficient vector: hiding, binding.
- Opening at z is an inner product with a public vector.
- The problem: convince the verifier of $\langle \vec{a}, \vec{b} \rangle = v$ in fewer than d group elements, without sending $\vec{a}$.

The inner-product argument: one fold

Split every vector in half and publish two cross terms:

$$L := \langle \vec{a}_{\text{lo}}, \vec{G}_{\text{hi}} \rangle + [\ell] H + [\langle \vec{a}_{\text{lo}}, \vec{b}_{\text{hi}} \rangle] U,$$

$$R := \langle \vec{a}_{\text{hi}}, \vec{G}_{\text{lo}} \rangle + [r_R] H + [\langle \vec{a}_{\text{hi}}, \vec{b}_{\text{lo}} \rangle] U.$$

Challenge u ; both sides fold:

$$\vec{a}' := u \vec{a}_{\text{lo}} + u^{-1} \vec{a}_{\text{hi}}, \quad \vec{G}' := u^{-1} \vec{G}_{\text{lo}} + u \vec{G}_{\text{hi}}, \quad \vec{b}' := u^{-1} \vec{b}_{\text{lo}} + u \vec{b}_{\text{hi}}$$

$$P + [v] U + [u^2] L + [u^{-2}] R = \langle \vec{a}', \vec{G}' \rangle + [r + u^2 \ell + u^{-2} r_R] H + [\langle \vec{a}', \vec{b}' \rangle] U$$

The same claim at half the length. The verifier folds $\vec{G}$ and $\vec{b}$ itself and never sees $\vec{a}$.

The recursion

After $k = \log_2 d$ rounds every vector has length 1. The proof: $2k$ points (L_j, R_j) , one scalar a , one blinder r^* .

$$P + [v] U + \sum_{j=1}^k ([u_j^2] L_j + [u_j^{-2}] R_j) \stackrel{?}{=} [a] G^{(0)} + [r^*] H + [a \cdot b^{(0)}] U$$

- The scalar $b^{(0)} = \prod_j (u_j^{-1} + u_j z^{2^{j-1}})$, with u_1 the last challenge: logarithmic work.
- The point $G^{(0)} = \langle \vec{s}, \vec{G} \rangle$: one multi-scalar multiplication, shared across a batch of proofs.
- Binding: two openings would give a discrete-log relation among $\vec{G}, H$; an extractor rewinds three challenges per round.

The Orchard circuit

- Size: 2^{11} rows, 10 advice columns; arithmetic over $\mathbb{F}_p$ so Pallas points cm , rk , cv are native; commitments on Vesta.
- Gadgets: Sinsemilla (commitments, tree), Poseidon (nullifier), elliptic-curve arithmetic, lookups for range checks.
- The instance: $anchor$, cv , nf , rk , cmx , $flags$.
- Generators from a hash: no trusted setup, no ceremony.
- Proof size $2720 + 2272 a$ bytes for a Actions: 4992 for one.

Every clause of the statement is a set of rows; one proof, one random point, one inner-product argument.

The whole picture

check	Bitcoin	Ironwood
exists	UTXO lookup	Merkle path, inside the proof
not twice	delete the entry	nullifier, checked against the set
conserved	add the amounts	$\sum cv$, binding signature
authorised	script, signature	A6/A7, signature under rk

The validator learns that the four checks pass — and nothing about the note.